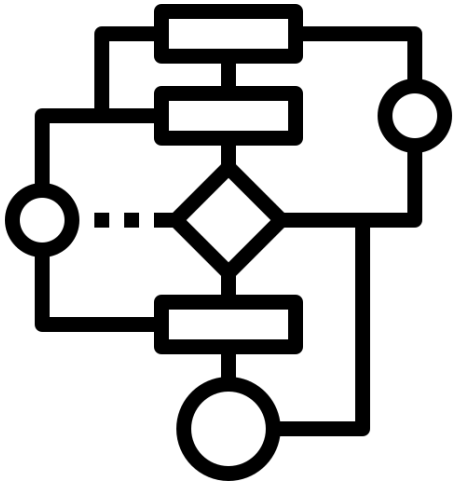
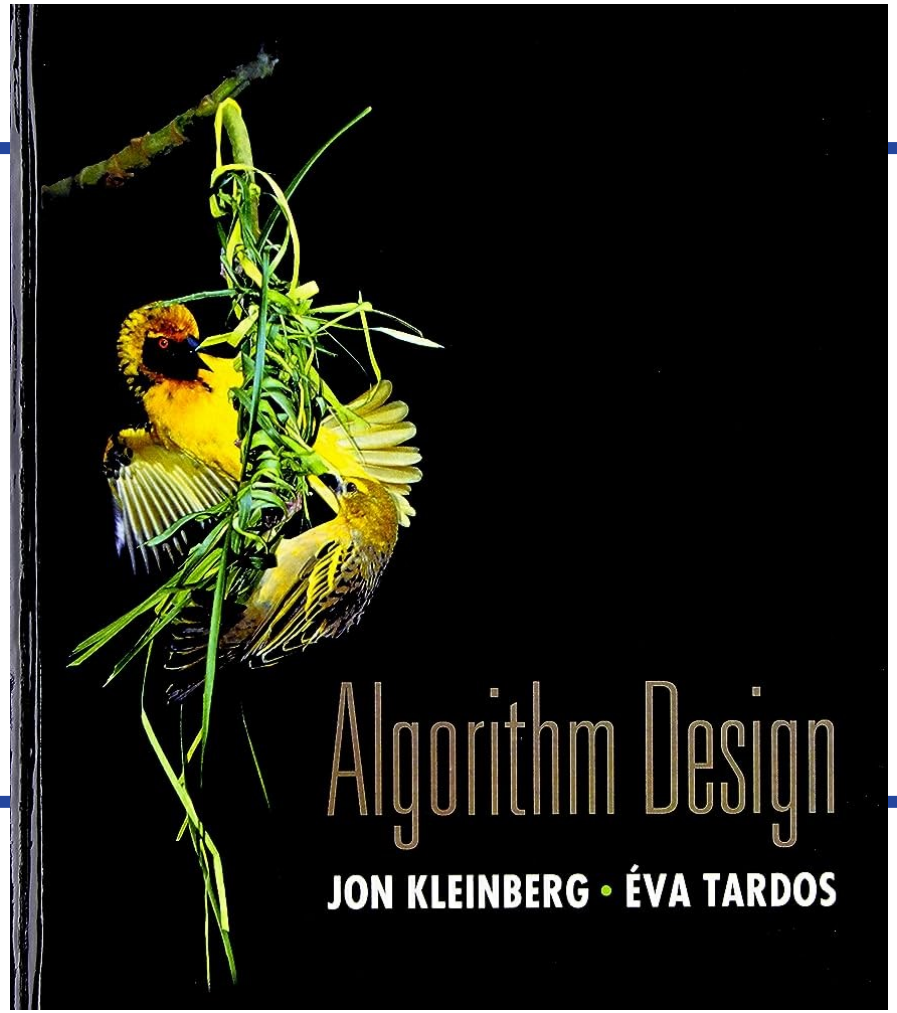


CS 310: Algorithms

Lecture 7

Instructor: Naveed Anwar Bhatti



Chapter 3:
Graphs

BFS: Live Poll 1

Consider a complete undirected graph where every vertex ***V*** has an edge with every other vertex. You are going to perform a Breadth-First Search (BFS) on this graph.

Which of the following expressions give equivalent time complexity in terms of the Big O notation of the BFS for this graph?

A. $V^2 = \mathbf{O(V^2)}$ ✓

B. 2E = $V(V-1) = V^2 - V = \mathbf{O(V^2)}$ ✓ 😈

C. V+2E = $V + V(V-1) = V^2 = \mathbf{O(V^2)}$ ✓ 😈

D. V+E = $V + (V(V-1))/2 = V + (V^2 - V)/2 = \mathbf{O(V^2)}$ ✓ 😈

E. All of Above

F. None of Above

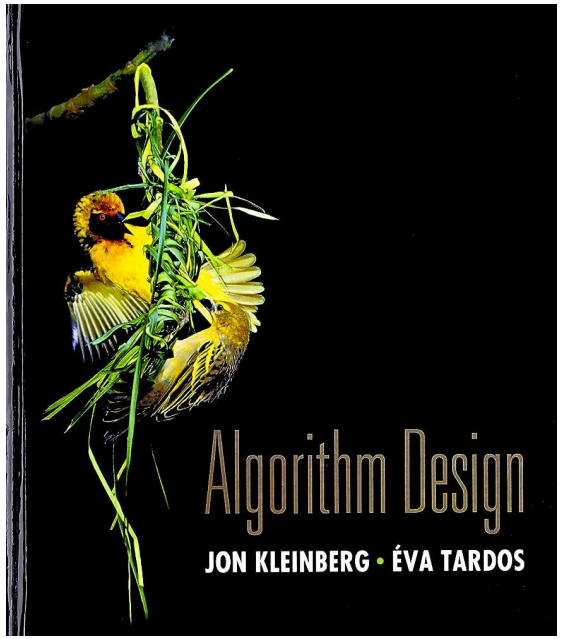
Breadth First Search: Analysis

Theorem: The above implementation of BFS runs in O(m + n) time if the graph is given by its adjacency representation.

- **Proof:**
  - Easy to prove **$O(n^2)$** running time:
    - at most n lists L[i]
    - each node occurs on at most one list; for loop runs $\leq$ n times
    - when we consider node u, there are $\leq$ n incident edges (u, v), and we spend O(1) processing each edge
  - Actually, runs in **O(m + n)** time:
    - when we consider node u, there are deg(u) incident edges (u, v)
    - total time processing edges is $\Sigma_{u \in V}$ deg(u) = 2m ▪

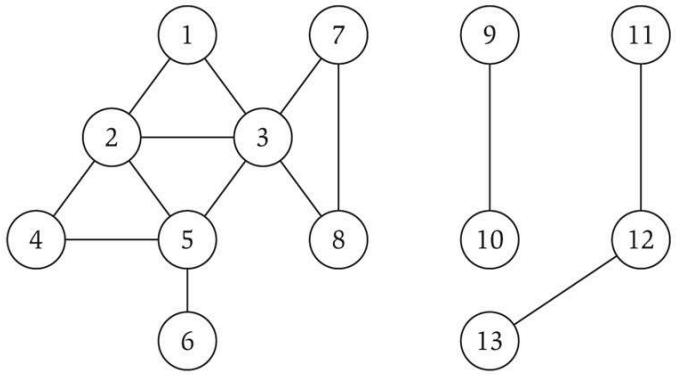
each edge (u, v) is counted exactly twice in sum: once in deg(u) and once in deg(v)

Read the book



Connected Component

- Connected component. Find all nodes reachable from s.

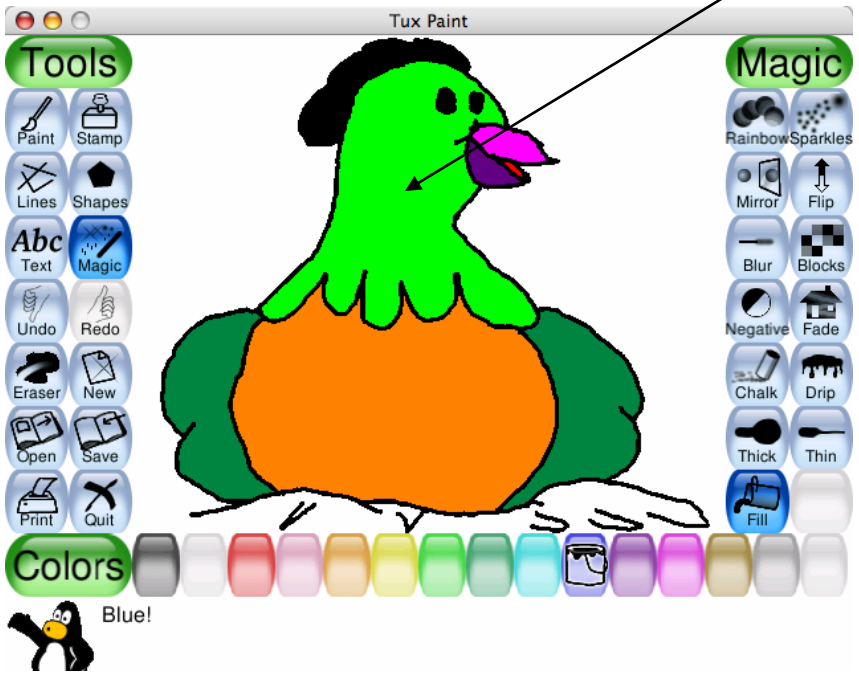


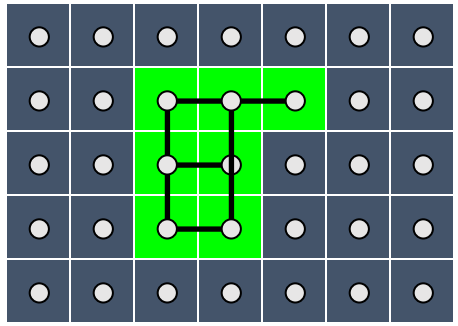
- Connected component containing node 1 = { 1, 2, 3, 4, 5, 6, 7, 8 }.

Flood Fill

- Flood fill. Given lime green pixel in an image, change color of entire blob of neighboring lime pixels to blue.
  - Node: pixel.
  - Edge: two neighboring lime pixels.
  - Blob: connected component of lime pixels.

recolor lime green blob to blue





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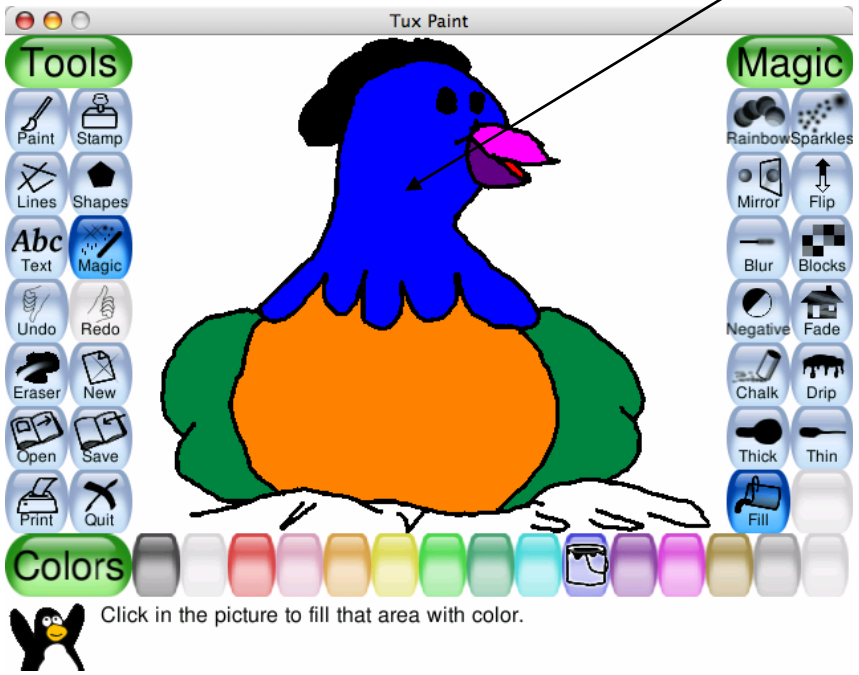
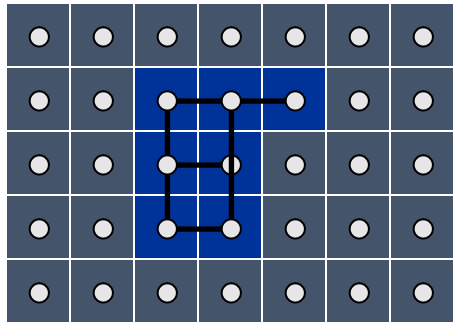
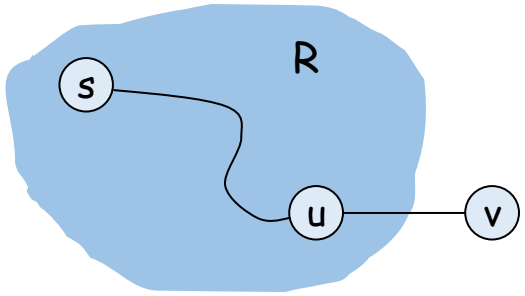




Connected Component

- Connected component. Find all nodes reachable from s.

```
R will consist of nodes to which s has a path
Initially R = {s}
While there is an edge (u, v) where u ∈ R and v ∉ R
  Add v to R
Endwhile
```



it's safe to add v

- Theorem. Upon termination, R is the connected component containing s.
  - BFS = explore in order of distance from s.
  - DFS = explore in a different way.

Section 3.4:
Testing Bipartiteness

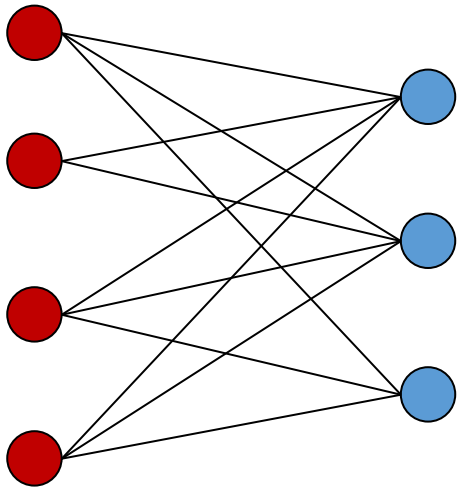
Administrivia

- **Assignment 1 – Released**
- **Quiz 2 on Wednesday 27th**

Bipartite Graphs

- **Def:** An undirected graph G = (V, E) is bipartite if the nodes can be colored red or blue such that every edge has one red and one blue end.

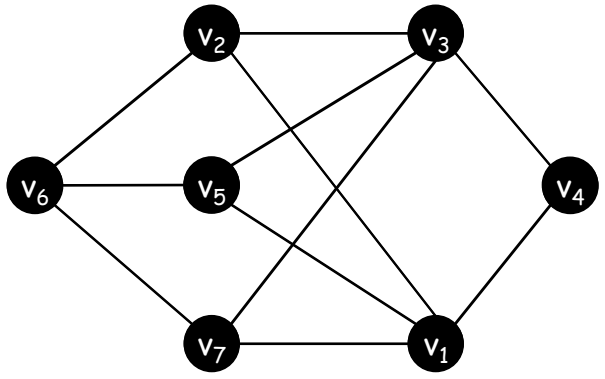
- Applications.
  - Stable matching: courses = blues, TAs = red.
  - Scheduling: jobs = blue, machines = red



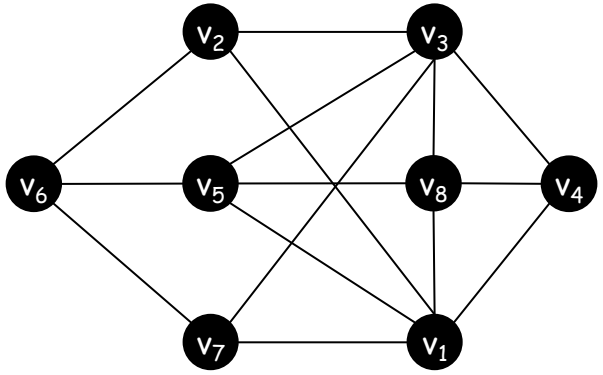
a bipartite graph

Testing Bipartiteness

Given a graph **G1** and **G2**, which one is bipartite?

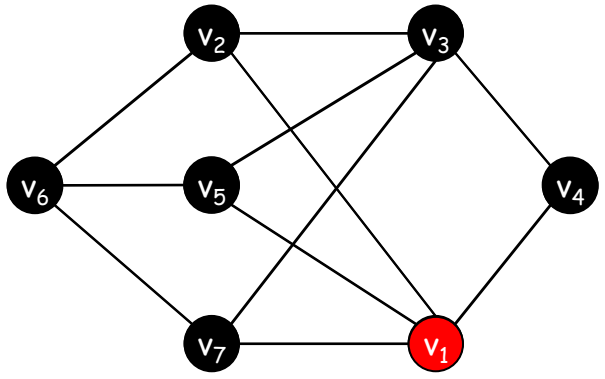


G1

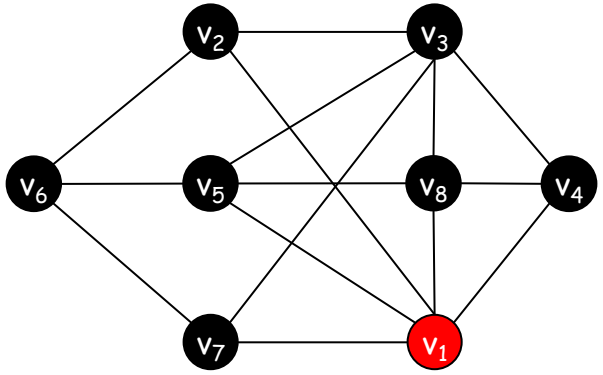


G2

Testing Bipartiteness



G1



G2

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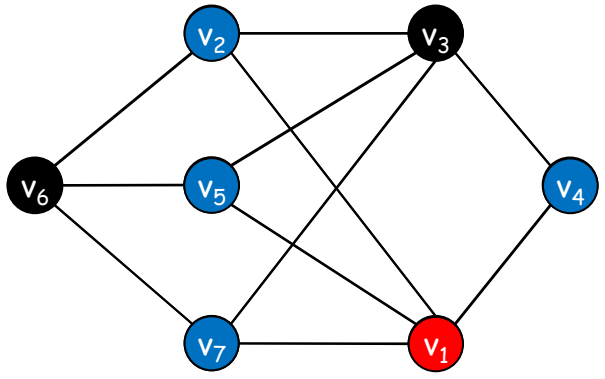

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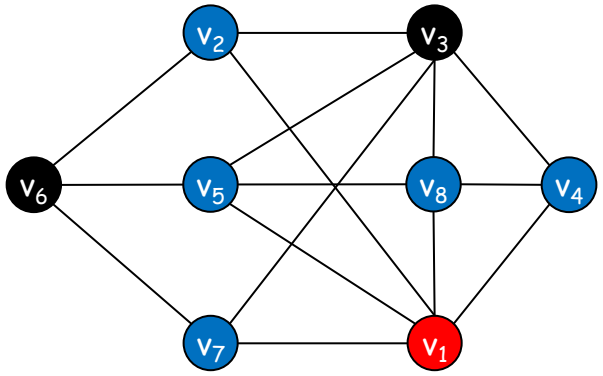
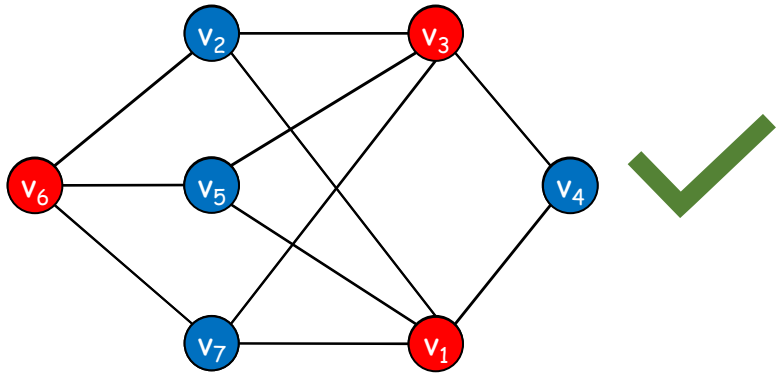


G2

Testing Bipartiteness



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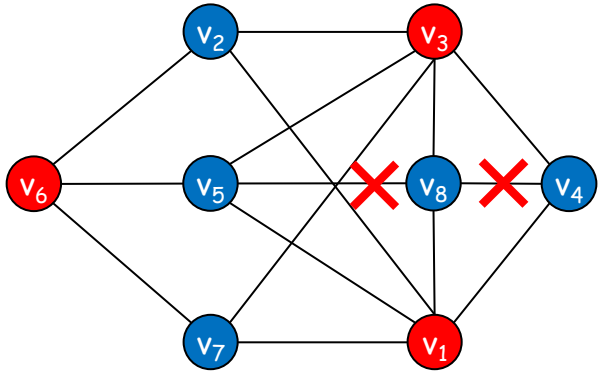
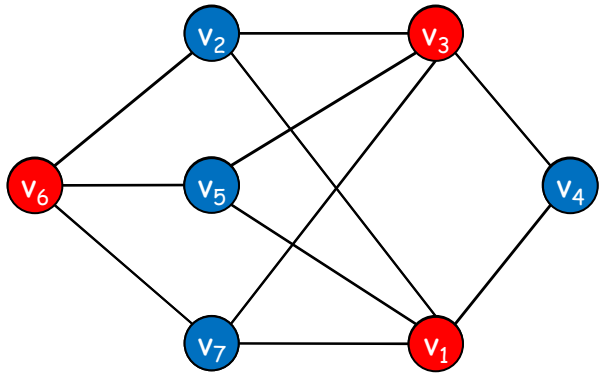


G2

Testing Bipartiteness



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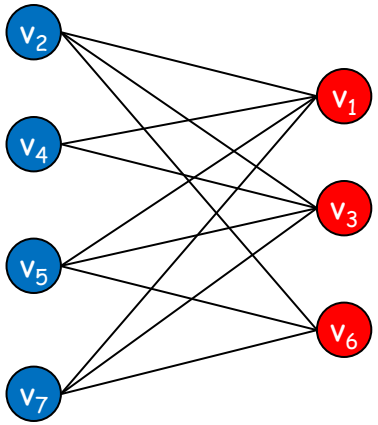
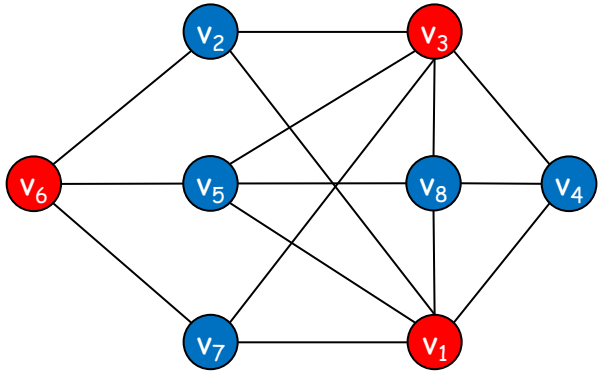


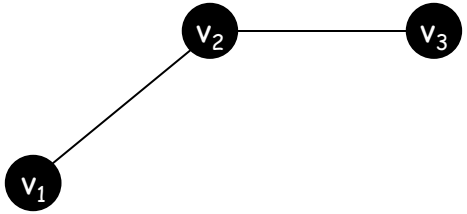
another drawing of G1



G2

Testing Bipartiteness

Before attempting to design an algorithm, we need to understand structure of bipartite graphs



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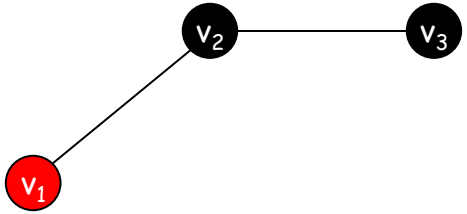

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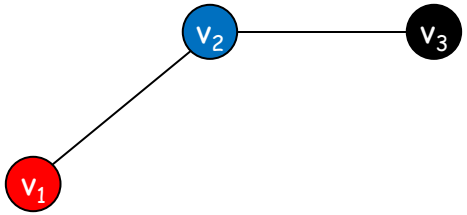

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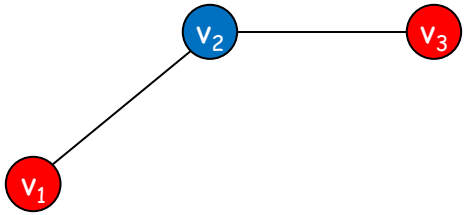

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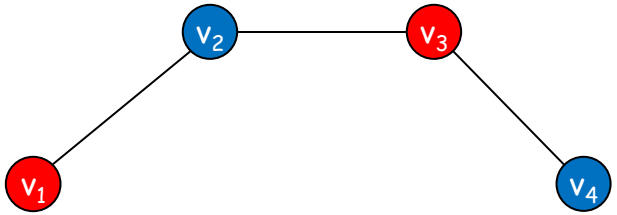

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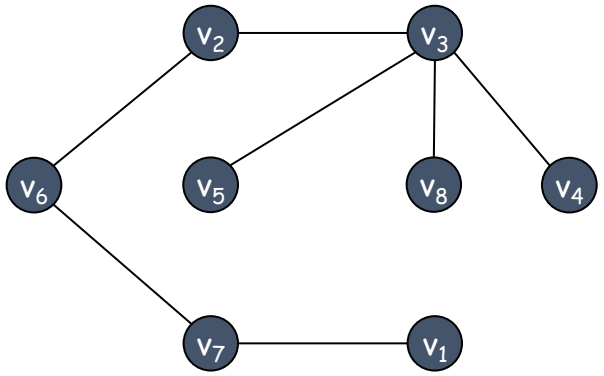

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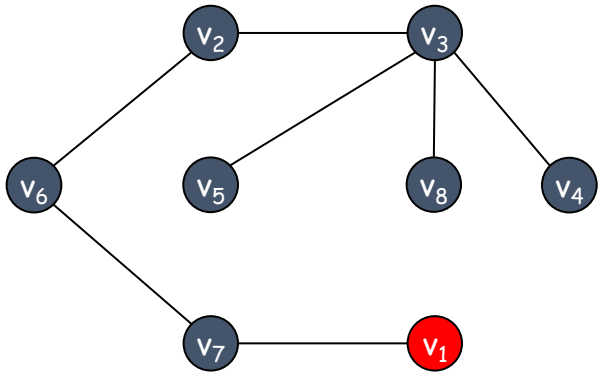

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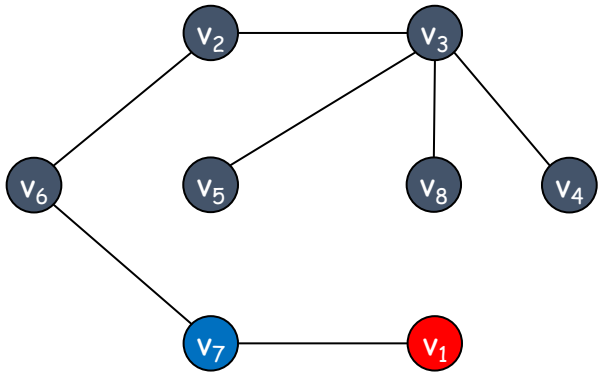

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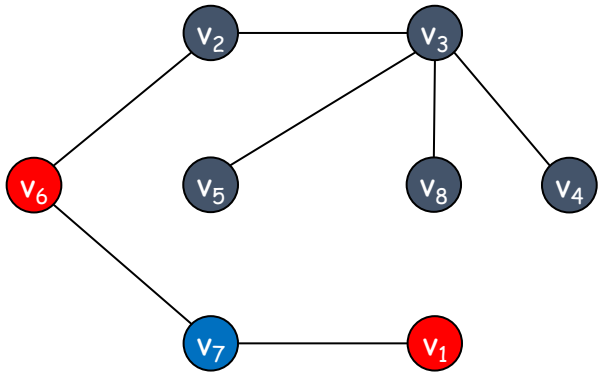

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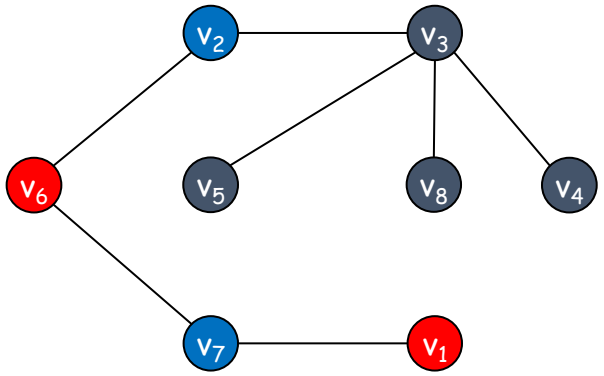

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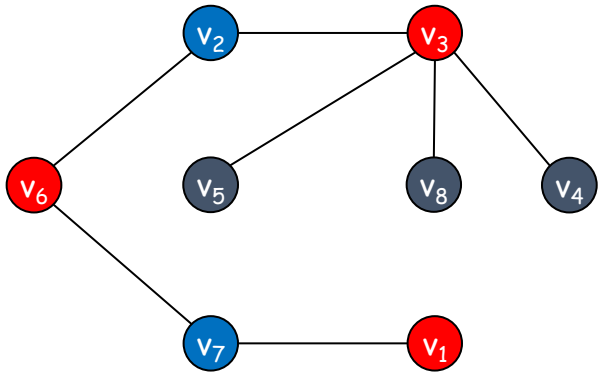

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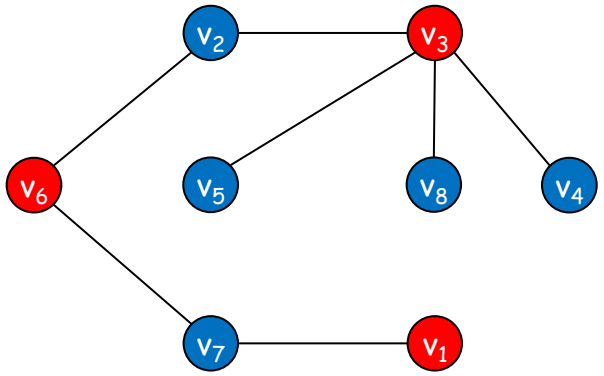
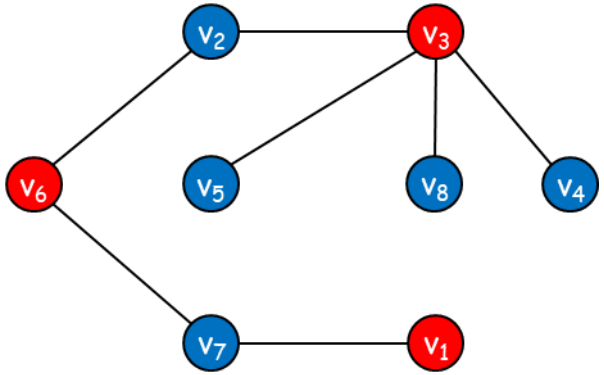
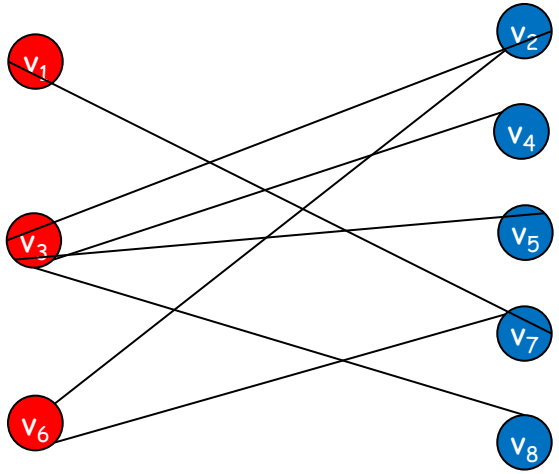


Testing Bipartiteness

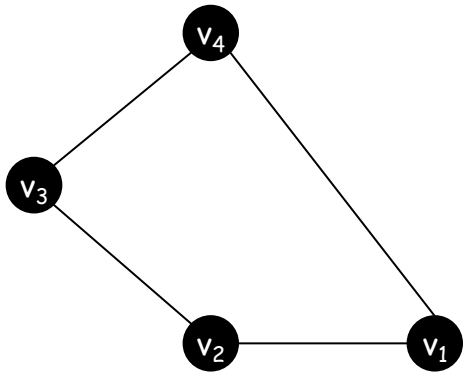
Observation: *A* graph ***G*** without cycles is always **bipartite**.





Testing Bipartiteness

What about Cycles? Let explore **even cycles**



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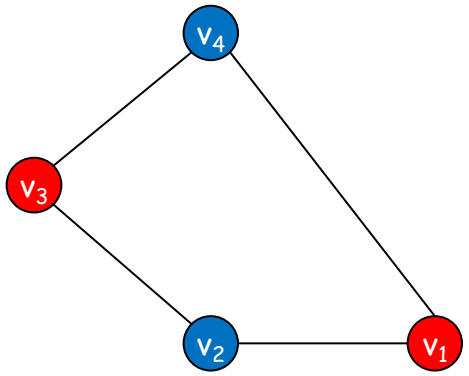

Testing Bipartiteness

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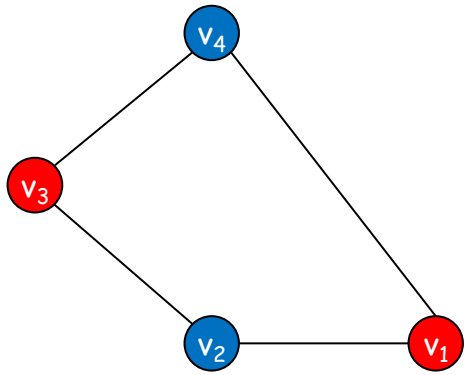

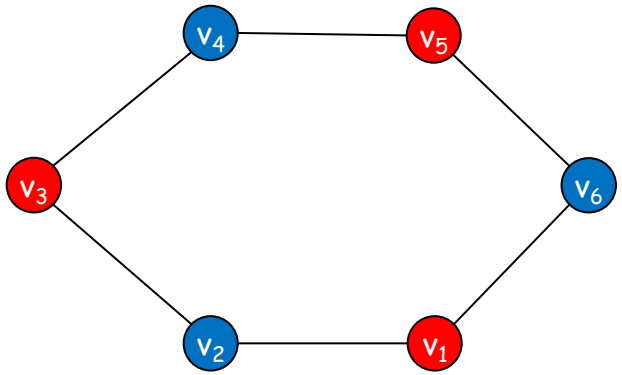

Testing Bipartiteness

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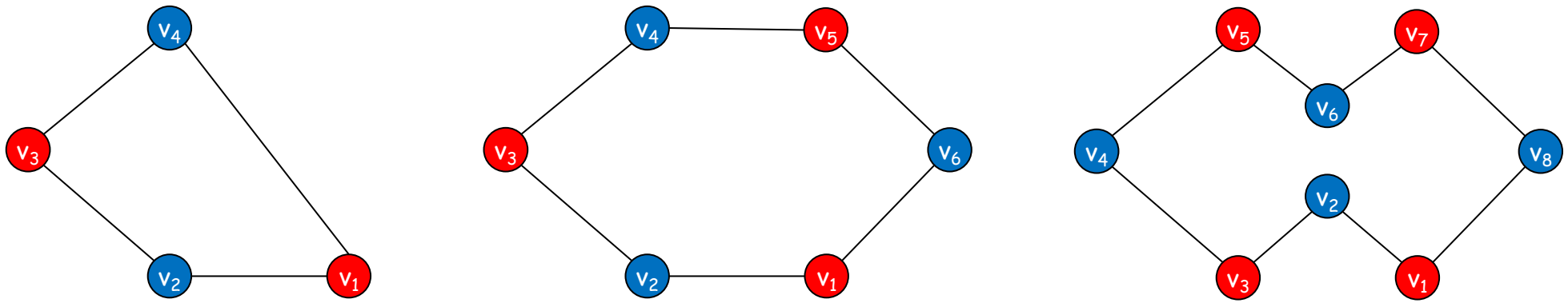
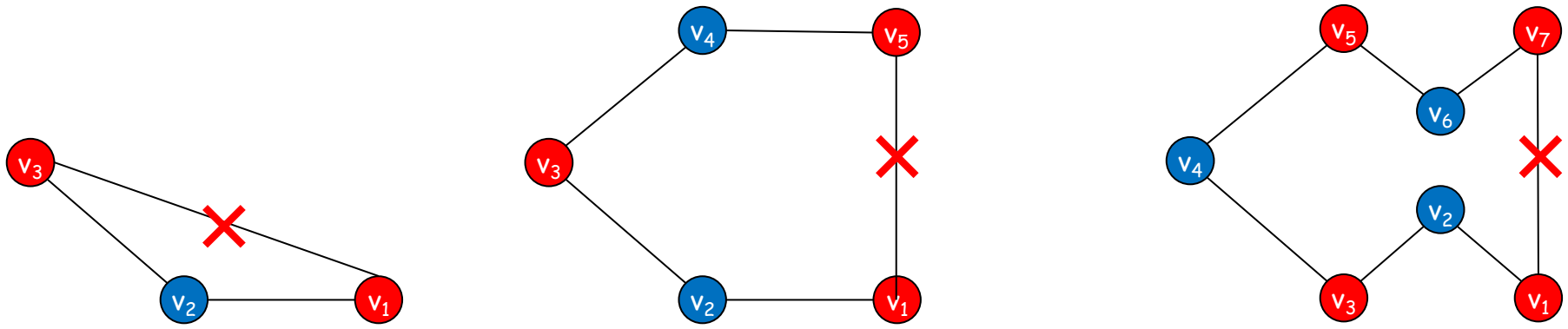


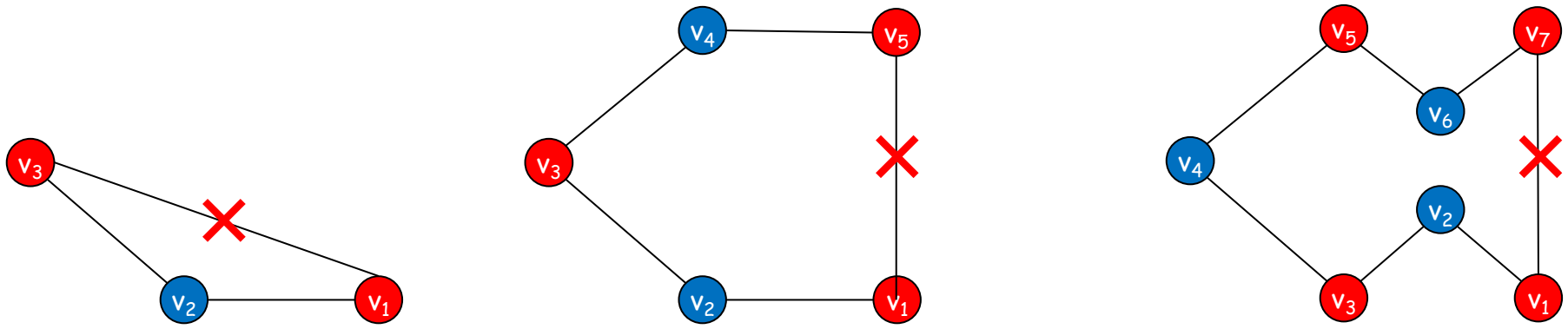
Testing Bipartiteness

What about **odd cycles?**

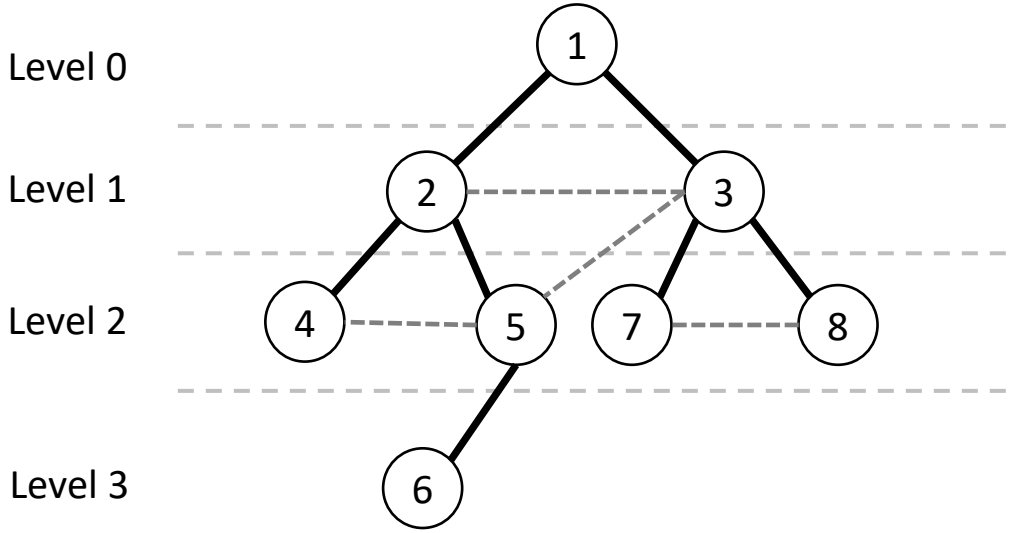


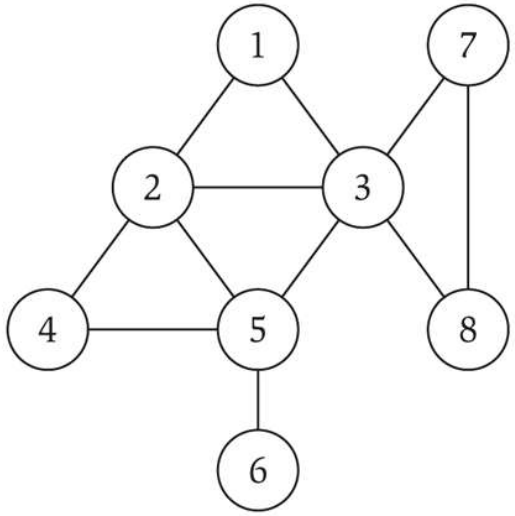
Testing Bipartiteness

Lemma 1: If a graph **G** is bipartite, it cannot contain an odd length cycle.

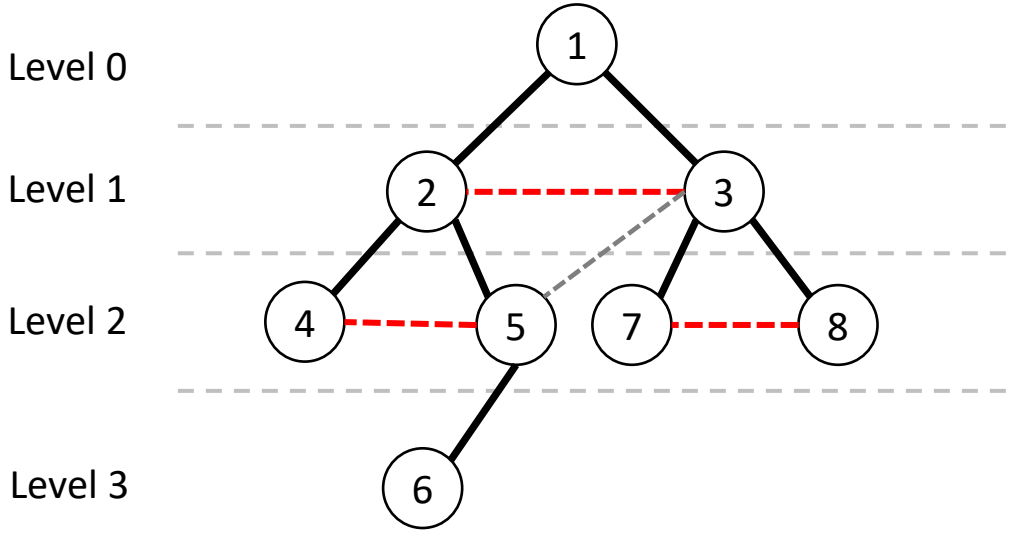


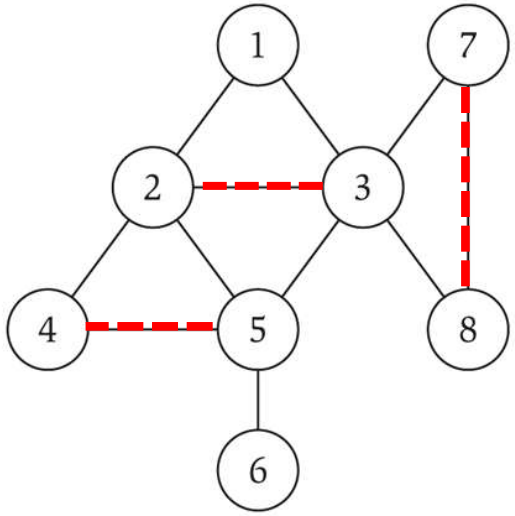
Bipartiteness and BFS



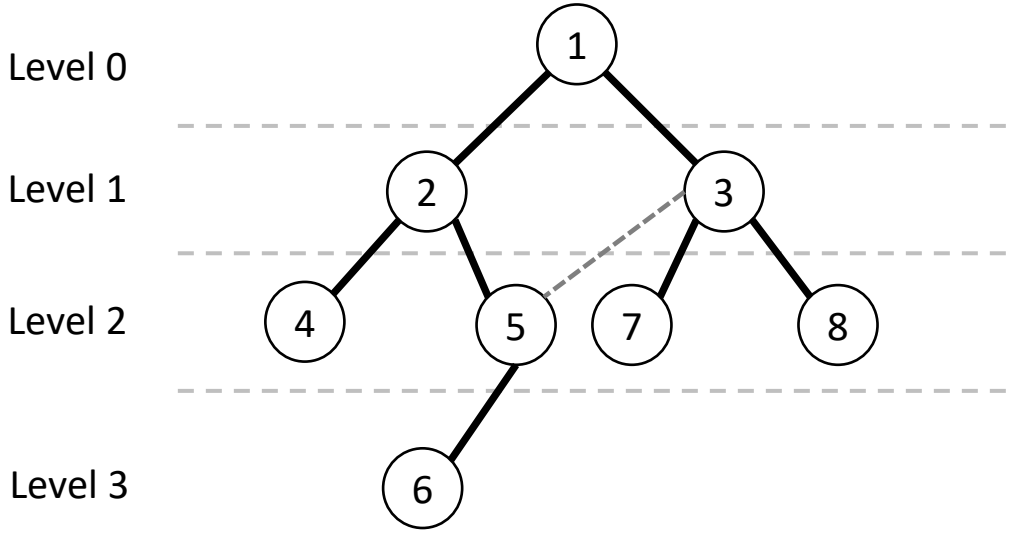


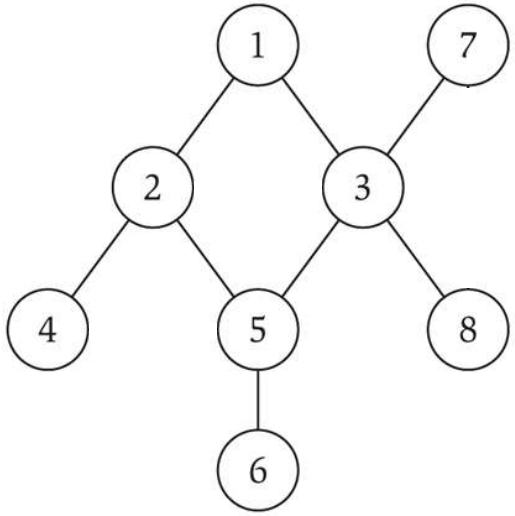
Bipartiteness and BFS





Bipartiteness and BFS



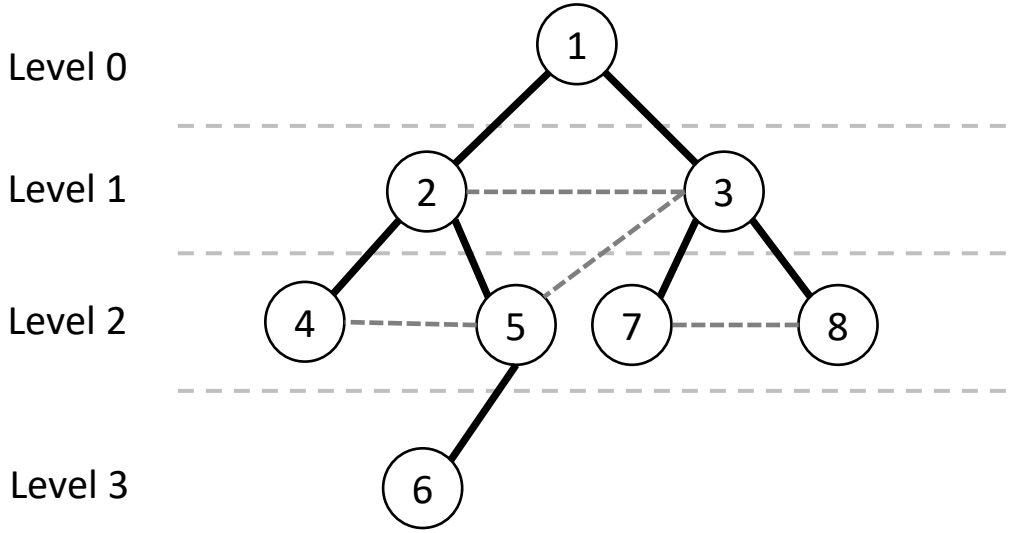


Bipartite Graphs and BFS

Hence…

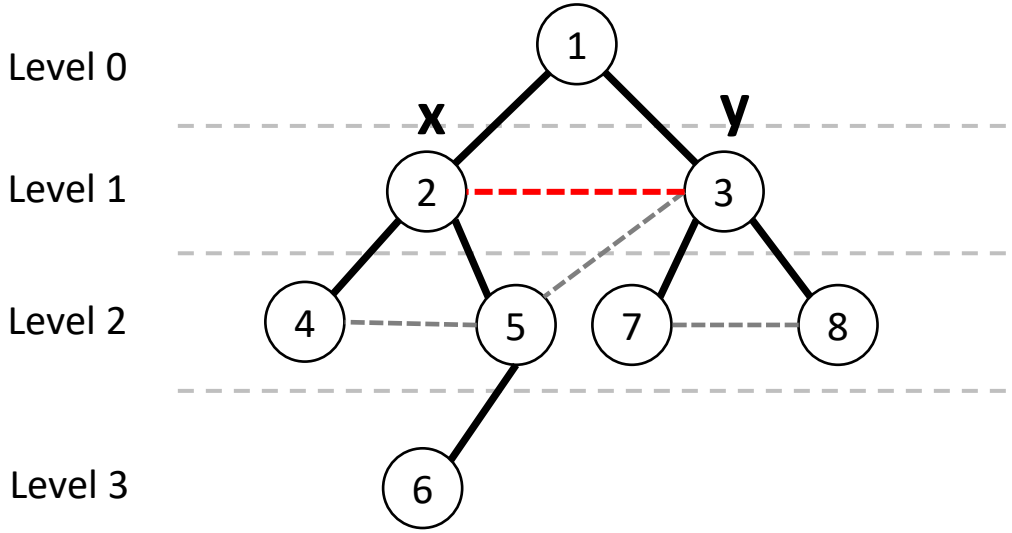
- **Lemma 2**. Let ***G*** be a connected graph, and let $\mathbf{L_0}$, …, $\mathbf{L_k}$ be the layers produced by **BFS** starting at node ***s***. Exactly one of the following holds.
  - (i) No edge of ***G*** joins two nodes of the same layer, and ***G*** is bipartite.
  - (ii) An edge of ***G*** joins two nodes of the same layer, and ***G*** contains an odd-length cycle (and hence is not bipartite).

Bipartiteness and BFS



Proof:

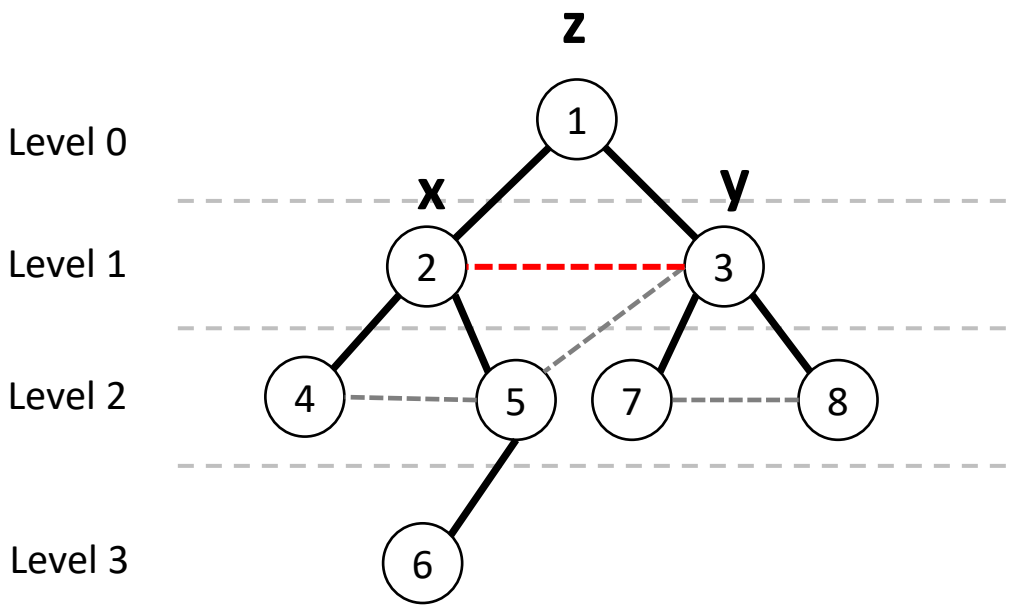
Bipartiteness and BFS



Proof:

- Suppose **(x, y)** is an edge with x, y in same level $\mathbf{L_j}$

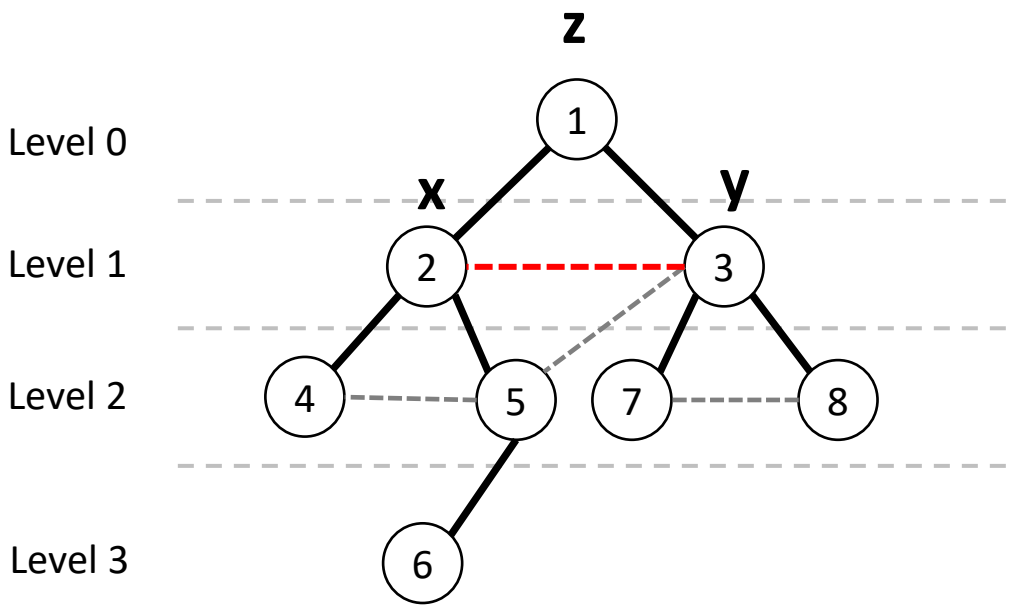
Bipartiteness and BFS



Proof:

- Suppose **(x, y)** is an edge with x, y in same level $\mathbf{L_j}$.
- Let **z** = lca(x, y) = lowest common ancestor.

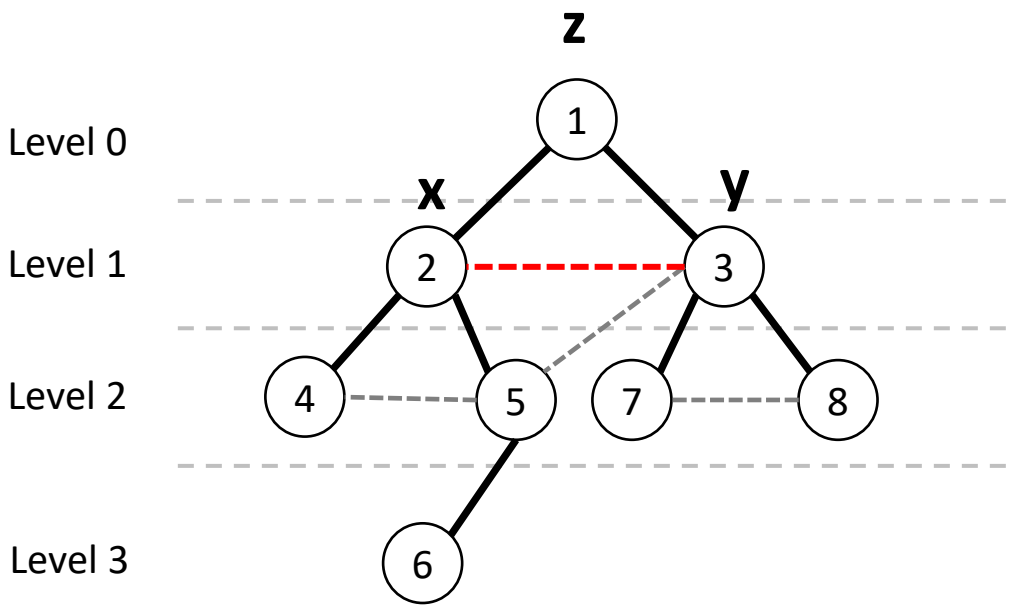
Bipartiteness and BFS



Proof:

- Suppose **(x, y)** is an edge with x, y in same level $\mathbf{L_j}$.
- Let **z** = lca(x, y) = lowest common ancestor.
- Let $\mathbf{L_i}$ be level containing **z**

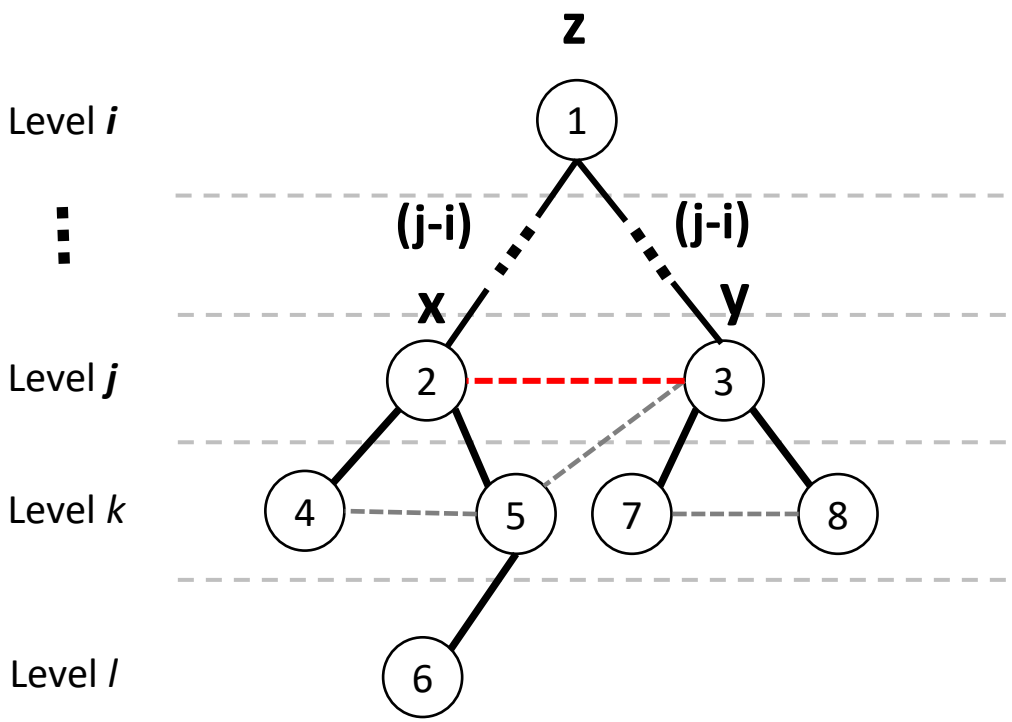
Bipartiteness and BFS



Proof:

- Suppose **(x, y)** is an edge with x, y in same level $\mathbf{L_j}$.
- Let **z** = lca(x, y) = lowest common ancestor.
- Let $\mathbf{L_i}$ be level containing **z**
- Consider cycle that takes edge from **x to y**, then path from **y to z**, then path from **z to x**.

Bipartiteness and BFS



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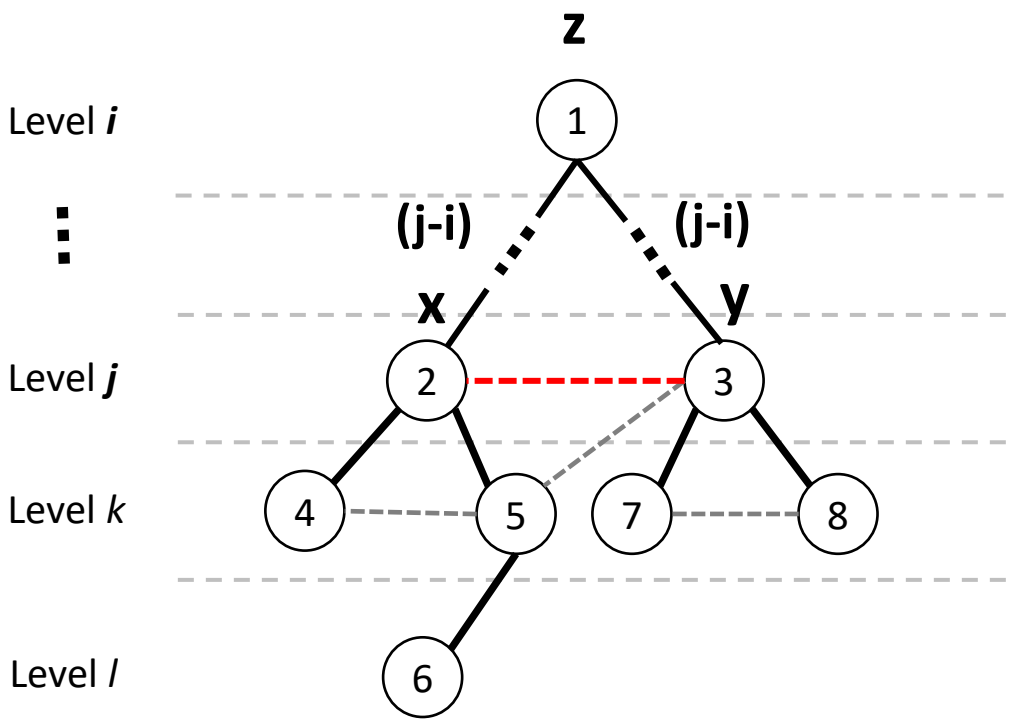
Bipartiteness and BFS



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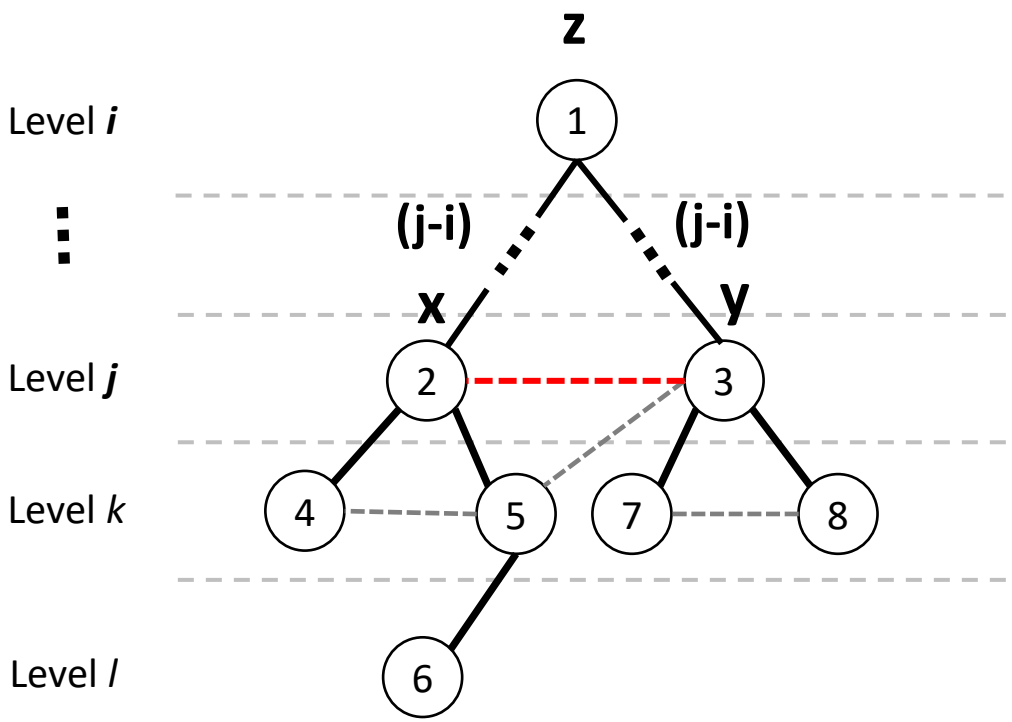
- Its length is **1** + **(j-i)** + **(j-i)**.

Bipartiteness and BFS



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- Its length is **1** + **(j-i)** + **(j-i)**.

$$\mathbf{1} + \underbrace{\mathbf{2\,(j\text{-}i)}}_{\text{Any number multiplied by 2 is even}}$$

Which always gives **Odd** length

Bipartite Graphs

Corollary (Based on *Lemma 1* and *2*)**:** A graph **G** is bipartite ***iff*** it contain no odd length cycle.

Testing Bipartite Graphs – Designing the Algorithm

```
procedure BFS(G,s)

    for each vertex v ∈ V[G] do
        explored[v] ← false
        d[v] ← ∞
    end for
    explored[s] ← true
    d[s] ← 0
    Q:= a queue data structure, initialized with s
    while Q ≠ ϕ do
        u ← remove vertex from the front of Q
        for each v adjacent to u do
            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```

Replace it with **color[v] = Black**

Replace it with **color[s] = Red**

color it with the opposite color of ‘u’

Add new ***if*** block. If **v** is already explored and has the same color as **u**, then the graph is not bipartite. Exit.

Testing Bipartite Graphs – Designing the Algorithm

```
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            if not explored[v] then
                explored[v] ← true
                d[v] ← d[u] + 1
                insert v to the end of Q
            end if
        end for
    end while

end procedure
```

Same Time Complexity

Thanks a lot

If you are taking a Nap, **wake up**.....Lecture Over